```
!pip install -q faiss-cpu transformers torch pandas

from google.colab import files
uploaded = files.upload()

import pandas as pd
file_name = list(uploaded.keys())[0]
df = pd.read_csv(file_name)

print("Dataset loaded successfully")
print(df.head())

TEXT_COLUMN = df.columns[0]
item_texts = df[TEXT_COLUMN].astype(str).tolist()

print("\nTotal items:", len(item_texts))
print("Sample item:", item_texts[0])

from transformers import AutoTokenizer, AutoModel
import torch
import numpy as np
import faiss

device = "cuda" if torch.cuda.is_available() else "cpu"

model_name = "sentence-transformers/all-MiniLM-L6-v2"

tokenizer = AutoTokenizer.from_pretrained(model_name)
encoder = AutoModel.from_pretrained(model_name).to(device)
encoder.eval()

def encode_texts(texts, batch_size=16):
    all_embeddings = []
    with torch.no_grad():
        for i in range(0, len(texts), batch_size):
            batch = texts[i:i+batch_size]
            tokens = tokenizer(batch, padding=True, truncation=True,
return_tensors="pt").to(device)
            outputs = encoder(**tokens)
            embeddings = outputs.last_hidden_state.mean(dim=1)
            embeddings = embeddings.cpu().numpy().astype("float32")
            all_embeddings.append(embeddings)
    return np.vstack(all_embeddings)

item_embeddings = encode_texts(item_texts)

faiss.normalize_L2(item_embeddings)

print("\nEmbeddings shape:", item_embeddings.shape)

embedding_dim = item_embeddings.shape[1]
```

```python
index = faiss.IndexFlatIP(embedding_dim)
index.add(item_embeddings)

print("FAISS index built with", index.ntotal, "items")

def get_similar_items(item_id, k=5):
    query_vector = item_embeddings[item_id:item_id+1]
    scores, indices = index.search(query_vector, k)
    return scores[0], indices[0]

query_id = 0
scores, idxs = get_similar_items(query_id, k=5)

print("\nQUERY ITEM:")
print(item_texts[query_id])

print("\nSIMILAR ITEMS:")

for score, idx in zip(scores, idxs):
    print(f"-> {item_texts[idx]} (score={score:.3f})")
```

```
<IPython.core.display.HTML object>
```

```
Saving sample_test.csv to sample_test.csv
Dataset loaded successfully
   sample_id                                    catalog_content  \
0     217392  Item Name: Gift Basket Village Gourmet Meat an...
1     209156  Item Name: NPG Dried Lotus Seeds 16 Oz, Uncook...
2     262333  Item Name: Annies Homegrown Macaroni and Chees...
3     295979  Item Name: Bear Creek Country Kitchens Creamy ...
4      50604  Item Name: Japanese Kelp Kombu Umami Soup Stoc...

                                          image_link
0  https://m.media-amazon.com/images/I/91GB1wC6Ob...
1  https://m.media-amazon.com/images/I/81VnzF1vkv...
2  https://m.media-amazon.com/images/I/51aCDMHMnI...
3  https://m.media-amazon.com/images/I/71dzRyLGPi...
4  https://m.media-amazon.com/images/I/71Yu21cGwr...
```

```
Total items: 100
Sample item: 217392
```


```
{"model_id":"bc9f6676fdd34fa1a6299744ae6e6d43","version_major":2,"version_minor":0}
```

```
{"model_id":"fa1e368e6da249cba23cdb9fbf952767","version_major":2,"version_minor":0}
```

```
{"model_id":"2e56ada486ae4ff8a9393d8d28307a5c","version_major":2,"version_minor":0}
```

```
{"model_id":"1acfd524dc404e92bf0b0f4536fe0548","version_major":2,"vers
ion_minor":0}

{"model_id":"8aa9ca07243a47198af6a2aa3bbf4f9e","version_major":2,"vers
ion_minor":0}

{"model_id":"0330dd8edccc42c8811786b3d349d348","version_major":2,"vers
ion_minor":0}

{"model_id":"69fdbb756f7d44cd8fd670efeabfb993","version_major":2,"vers
ion_minor":0}

BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different
task/architecture; not ok if you expect identical arch.


Embeddings shape: (100, 384)
FAISS index built with 100 items

QUERY ITEM:
217392

SIMILAR ITEMS:
-> 217392 (score=1.000)
-> 21764 (score=0.819)
-> 37027 (score=0.771)
-> 269532 (score=0.763)
-> 159533 (score=0.748)
```